Prepared by group 19

C programming quiz

FOP - CCA 1

Our Team

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Introduction

This project is a console-based Debugging Skills Quiz Game developed using C programming.

It is designed to improve logical thinking and error identification skills.

The game presents multiple-choice questions based on common C programming mistakes.

The user selects a difficulty level and answers questions to earn a score.

At the end, the program displays the final score along with performance remarks.

Objectives

- To understand basic C programming.
- To apply logical thinking.
- To create an interactive console game.

- Concept’s used:
- printf() and scanf()
- If-Else statements
- Loops
- Integer variables
- Score calculation

Algorithm

1. Start the program
2. Select dificulty level
3. Display quiz questions
4. Take user answer
5. Increase score if correct
6. Repeat for all questions
7. Display final score
8. End program

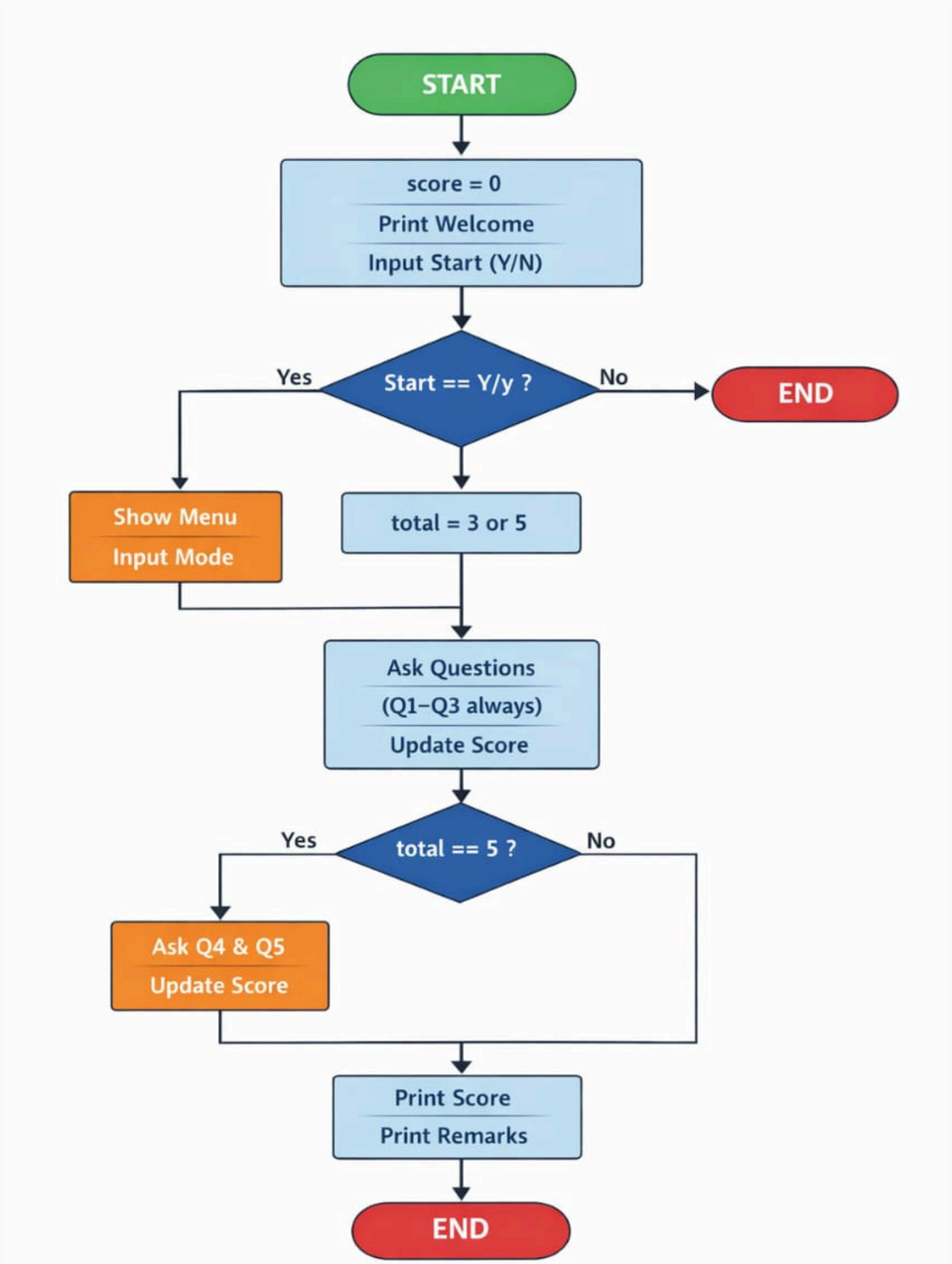
```c
#include <stdio.h>

int main() {
    int score = 0, choice, ans, total;
    char start;

    printf("=== Debugging Quiz Game ===\n");
    printf("Start game? (Y/N): ");
    scanf(" %c", &start);
    if(start!='Y' && start!='y') return 0;

    printf("\n1. Easy (3 Q)\n");
    printf("2. Hard (5 Q)\n");
    printf("Choose: ");
    scanf("%d", &choice);
```

Flowchart


START
score = 0
Print Welcome
Input Start (Y/N)
Start == Y/y ?
Yes
No
END
Show Menu
Input Mode
total = 3 or 5
Ask Questions
(Q1–Q3 always)
Update Score
total == 5 ?
Yes
No
Ask Q4 & Q5
Update Score
Print Score
Print Remarks
END

Code:

```c
#include <stdio.h>

int main() {
    int score = 0, choice, ans, total;
    char start;

    printf("=== Debugging Quiz Game ===\n");
    printf("Start game? (Y/N): ");
    scanf(" %c", &start);
    if(start!='Y' && start!='y') return 0;

    printf("\n1. Easy (3 Q)\n");
    printf("2. Hard (5 Q)\n");
    printf("Choose: ");
    scanf("%d", &choice);

    total = (choice==1)?3:5;

    /* Q1 */
    printf("\nQ1: scanf(\"%%d\", x); What's wrong?\n");
    printf("1. x should be float\n");
    printf("2. Use %%f\n");
    printf("3. Use &x\n");
    printf("4. Remove printf\n");
    scanf("%d",&ans);
    score += (ans==3)?1:-1;

    /* Q2 */
    printf("\nQ2: int a=5,b=2; a/b = ?\n");
    printf("1. 2.5\n");
    printf("2. 2.7\n");
    printf("3. 3.1\n");
    printf("4. 2.2\n");
    scanf("%d",&ans);
    score += (ans==1)?1:-1;

    /* Q3 */
    printf("\nQ3: for(i=1; ? ; i++) prints 1-5\n");
    printf("1. i < 5\n");
    printf("2. i <= 5\n");
    printf("3. i > 5\n");
    printf("4. i >= 5\n");
    scanf("%d",&ans);
    score += (ans==2)?1:-1;

    if(total==5){

        /* Q4 */
        printf("\nQ4: if(x=5) mistake?\n");
        printf("1. x should be float\n");
        printf("2. Initialize to 5\n");
        printf("3. Use == instead of =\n");
        printf("4. Use %%d\n");
        scanf("%d",&ans);
        score += (ans==3)?1:-1;
```

Code:

```
        /* Q5 */
        printf("\nQ5: int arr[3]; arr[3]?\n");
        printf("1. Size should be 4\n");
        printf("2. Index out of bounds\n");
        printf("3. Use float\n");
        printf("4. Use %%f\n");
        scanf("%d",&ans);
        score += (ans==2)?1:-1;
    }

    printf("\nFinal Score: %d\n", score);

    if(score<=0) printf("Remarks: Can do better\n");
    else if(score<=2) printf("Remarks: Well done\n");
    else if(score<=4) printf("Remarks: Very good\n");
    else printf("Remarks: Mastered\n");

    return 0;
}
```

Output

```
=== Debugging Quiz Game ===
Start game? (Y/N): y

1. Easy (3 Q)
2. Hard (5 Q)
Choose: 1

Q1: scanf("%d", x); What's wrong?
1. x should be float
2. Use %f
3. Use &x
4. Remove printf
3

Q2: int a=5,b=2; a/b = ?
1. 2.5
2. 2.7
3. 3.1
4. 2.2
1

Q3: for(i=1; ? ; i++) prints 1-5
1. i < 5
2. i <= 5
3. i > 5
4. i >= 5
2

Final Score: 3
Remarks: Very good
```

Conclusion

- The project successfully implements a debugging quiz game.
- It strengthens logical thinking and error detection skills.
- Demonstrates practical use of conditional statements in C.
- Can be enhanced by adding more questions and levels.

Thank you